Proposal for Final Project

Project Title: Hospital Appointment System

1. Introduction

The Hospital Appointment System is a C++ console-based project designed to help patients book appointments with doctors easily.

This system will allow users to add doctors, view doctor availability, book appointments, and view scheduled appointments.

It will demonstrate the application of all basic C++ programming concepts studied in the Programming Fundamentals course.

2. Objectives

- To apply fundamental programming concepts in C++ practically.
- To design a real-world simple appointment system using console-based interaction.
- To learn how to use variables, arrays, functions, structures, loops, and conditional statements effectively.

3. Scope

This system will work for a small hospital or clinic where:

- Doctors can be added with their available timings.
- Patients can book appointments with available doctors.
- All appointments for a specific day can be viewed.
- Optional feature: Patients can cancel appointments.

4. Tools and Technologies

Programming Language: C++

Compiler: Code::Blocks / Dev C++ / Turbo C++ / Visual Studio

Platform: Console Application (Text-based, no graphics)

5. Features of the System

- Add Doctor Information
- Book Appointment for Patients
- View Doctor's Available Slots
- View All Scheduled Appointments
- Cancel Appointments (Optional)

6. Course Learning Outcomes (CLOs) Covered

(Details of CLO-01, CLO-02, CLO-03, CLO-04 covered through project components.)

7. System Design Overview

Data Structures:

struct Doctor { int id; string name; string specialization; string availableDays; };

struct Patient { int id; string name; string disease; string appointmentTime; int doctorId; };

Functions:

- addDoctor()
- bookAppointment()
- viewAppointments()
- cancelAppointment() (optional)

Menu Flow:

========== Hospital Appointment System ==========

1. Add Doctor

2. Book Appointment

3. View Appointments

4. Cancel Appointment

5. Exit

Enter your choice:

8. Expected Working

User interacts with menu to add doctors, book or cancel appointments, and view schedule until exiting.

9. Target Audience

- Beginner-level developers practicing programming concepts.
- Students learning basic C++ through real-life projects.
- Hospitals or clinics (if expanded later) can use this system.

10. Conclusion

The Hospital Appointment System project will help students practically understand how to:

- Apply all C++ basics (variables, structures, loops, functions, control structures).
- Develop real-world logic for simple business systems.
- Build and manage data using arrays and user-defined structures.

This project ensures full application of Programming Fundamentals course concepts.